ART LIBRARY SYSTEM

Presentation



Project Idea

- Manage art books and members
- Track borrowing process
- Organize DIY workshops
- Provide a structured library system



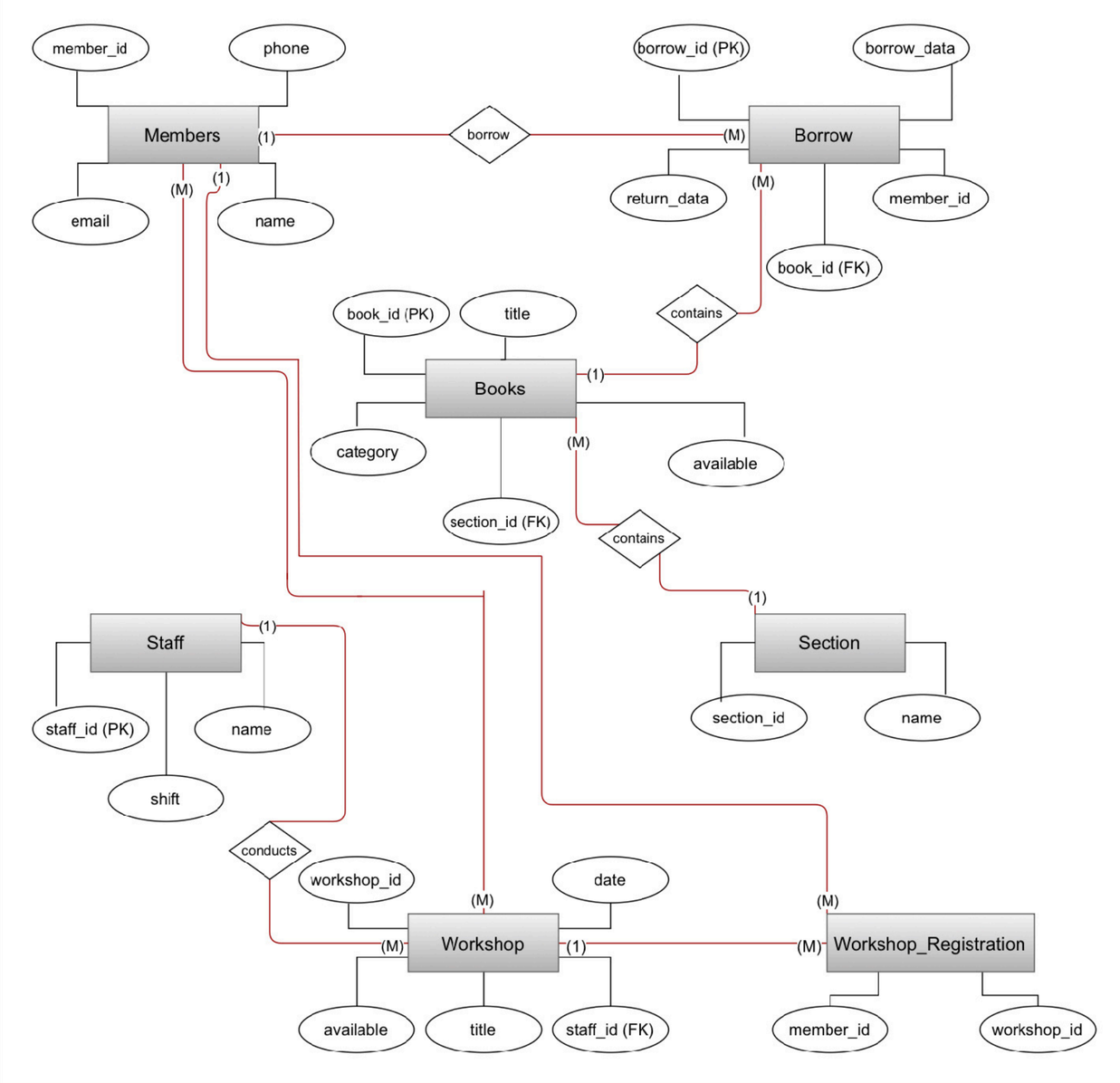
Entities

- Members
- Books
- Borrow
- Staff
- Workshop
- Section
- Workshop_Registration



ER Diagram

- Shows entities, attributes, and relationships
- Includes primary and foreign keys
- Represents the structure of the system



Relationships

- Member → Borrow (1:M)
- Book → Borrow (1:M)
- Section → Book (1:M)
- Staff → Workshop (1:M)
- Member ↔ Workshop (M:N)



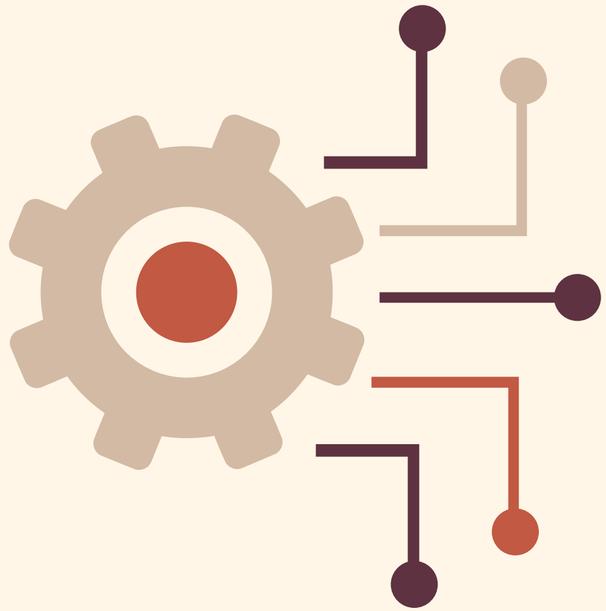
Special Features

- Best seller books (calculated using queries)
- Workshop registration system
- Organized book sections
- Easy tracking of borrowed books



Implementation

- Database created using MySQL
- Tables with primary and foreign keys
- Data inserted into each table
- Queries applied for data analysis



Database Tables Implementation

Query 1

Limit to 1000 rows

```sql
CREATE TABLE Book (
    book_id INT PRIMARY KEY,
    title VARCHAR(150) NOT NULL,
    category VARCHAR(100) NOT NULL,
    available VARCHAR(10) CHECK (available IN ('Yes', 'No')),
    section_id INT,
    FOREIGN KEY (section_id) REFERENCES Sections(section_id)
);

describe book;
```

Query 1

Limit to 1000 rows

```sql
CREATE TABLE workshop_registration (
    member_id INT,
    workshop_id INT,
    PRIMARY KEY (member_id, workshop_id),
    FOREIGN KEY (member_id) REFERENCES Members(member_id),
    FOREIGN KEY (workshop_id) REFERENCES Workshops(workshop_id)
);
describe workshop_registration;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Member_ID	int	NO	PRI	NULL	
Workshop_ID	int	NO	PRI	NULL	

Query 1

Limit to 1000 rows

```sql
CREATE TABLE Staff (
    Staff_ID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Shift VARCHAR(20) NOT NULL,
    Email VARCHAR(100) UNIQUE
);

describe staff;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Staff_ID	int	NO	PRI	NULL	
Name	varchar(100)	NO		NULL	
Shift	varchar(20)	NO		NULL	
Email	varbinary(100)	YES	UNI	NULL	

- An associative table used to handle the many-to-many relationship between members and workshops

SQL Queries Implementation

View

```sql
CREATE VIEW Borrow_Report AS
SELECT
    b.borrow_ID,
    m.name AS member_name,
    bk.title AS book_title,
    b.borrow_date,
    b.return_date,
    CASE
        WHEN b.return_date IS NULL THEN 'Not Returned'
        ELSE 'Returned'
    END AS status
FROM borrow b
LEFT JOIN members m
    ON b.member_ID = m.member_ID
LEFT JOIN book bk
    ON b.book_ID = bk.book_ID;

SELECT * FROM Borrow_Report;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

borrow_ID	member_name	book_title	borrow_date	return_date	status
1	Reem Mohammed	DIY Home Decoration	2026-04-01	2026-04-10	Returned
2	Abeer Saleh	Basic Watercolor Painting	2026-04-03	2026-04-12	Returned
3	Dana Sami	Crochet for Beginners	2026-04-05	NULL	Not Returned
4	Jana Fahad	Learn Art by Yourself	2026-04-07	NULL	Not Returned
5	Raghad Ali	Creative Handmade Gifts	2026-04-09	2026-04-16	Returned

Select

```sql
SELECT title, catagory
FROM Book
WHERE available = 'Yes';
```

Result Grid | Filter Rows:

title	catagory
DIY Home Decoration	DIY Crafts
Basic Watercolor Painting	Painting
Learn Art by Yourself	Self Learning

```sql
SELECT Name, Phone
FROM members
WHERE Name LIKE 'R%';
```

Result Grid | Filter Rows: | Expor

Name	Phone
Reem Mohammed	0501111111
Raghad Ali	0505555555

Join

```sql
SELECT book.Title, book.catagory, book.available
FROM book
JOIN section ON book.Section_ID =Section.Section_ID
WHERE Name = 'Best Sellers';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Title	catagory	available
DIY Home Decoration	DIY Crafts	Yes

Subquery

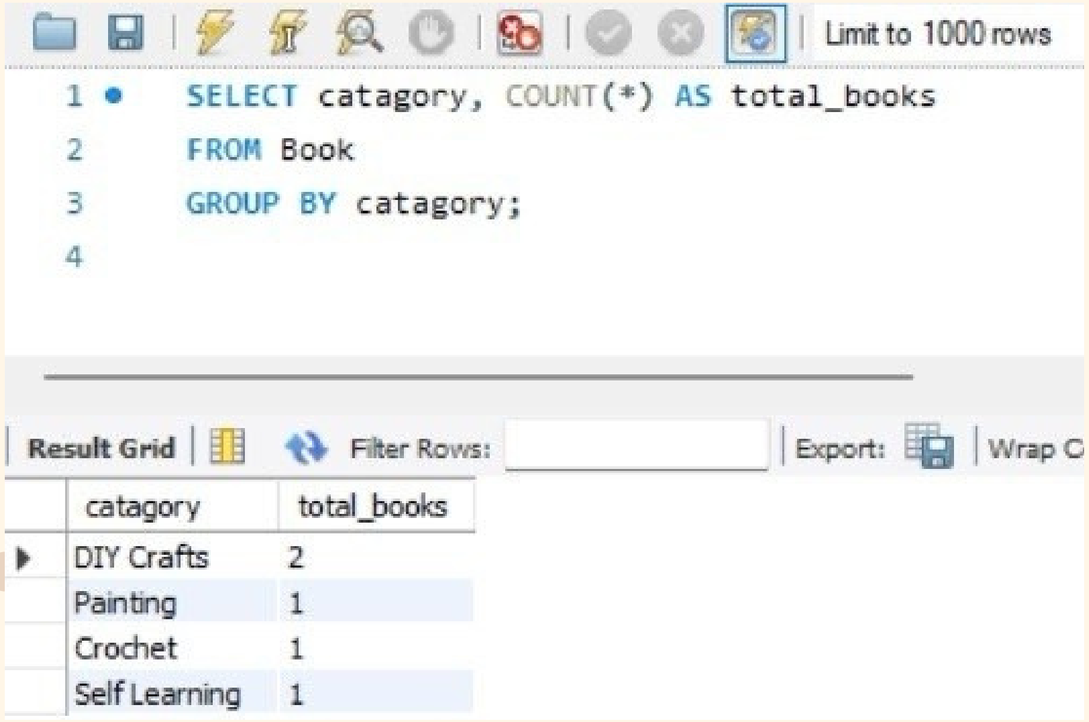
```sql
SELECT Name, phone
FROM members
WHERE Member_ID IN (
    SELECT Member_ID
    FROM borrow
    WHERE Return_Date IS NULL
);
```

Result Grid | Filter Rows: | Export:

Name	phone
Dana Sami	0503333333
Jana Fahad	0504444444

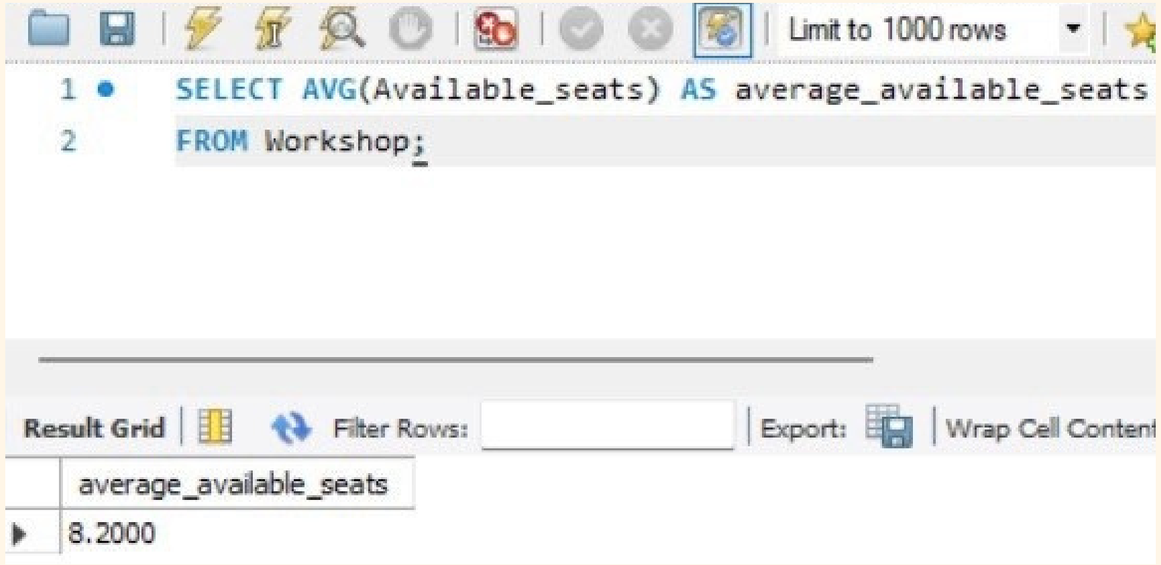
Retrieves members who have not returned their borrowed books

SQL Aggregate Queries (COUNT & AVG)



aggregate function COUNT

Counts the number of books
in each category



aggregate function AVG

Calculates the average
available seats in workshops

Conclusion

Efficient and organized system

This system organizes library data efficiently and helps manage books, members, and activities in a structured way

Supports real-world library operations

It reflects real-world library operations such as borrowing books and managing workshop activities

Easy to manage books and workshops

The system makes it easy to track books, manage workshop registrations, and handle library processes smoothly

THANK YOU

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